MESENCHYMAL STEM CELL-DERIVED EXTRACELLULAR VESICLES PROMOTE REPAIRING OF CORNEAL WOUND IN HUMAN CORNEA-ON-A-CHIP

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ABSTRACT

In this work, a microfluidic platform consisting of human corneal cells and a porous membrane was developed, replicating the multi-scale structural organization and biological phenotype of human cornea. Using this human cornea-on-a-chip, we develop an *in vitro* model of the corneal epithelial wound to study the cell-free therapy efficacy using extracellular vesicles (EVs).

KEYWORDS: Human Cornea-on-a-chip, Mesenchymal Stem Cells, Extracellular Vesicles, Wound Healing

INTRODUCTION

Mesenchymal stem cells (MSCs) with regenerative and differentiation capabilities have received much attention among ophthalmologists and scientists as an alternative modality for managing corneal diseases [1]. The paracrine effect of MSCs, mediated by EVs, has also been suggested for their therapeutic effect [1]. The cell-free nature of EVs has gained particular interest for their safety [2]. Organ-on-chips have attracted much attention to establishing standardized *in vitro* testing platforms in recent years. Organ-on-chips technology relies on our knowledge of human organs to engineer artificial constructs in which cells and their microenvironment are precisely controlled [3]. Here, the integrated human cornea-on-a-chip is used to evaluate MSC-EVs' efficacy for the corneal epithelial wound repair.

EXPERIMENTAL

The microdevice is fabricated by a glass substrate with two PDMS layers, and a collagen-coated membrane is sandwiched between the two PDMS layers (Figure 1A). Then, the immortalized human corneal epithelial (HCEpi) cells and endothelial (HCEnd) cells are introduced into the device through the upper and lower microchannels, respectively (Figure 1A). Once the epithelial layer is confluent, the medium is aspirated from the upper culture zone to create an air-liquid interface (ALI) and mimic the ocular surface. After 14 days, cell viability, trans-epithelial electrical resistance (TEER) and histological section are characterized to evaluate the corneal integrity.

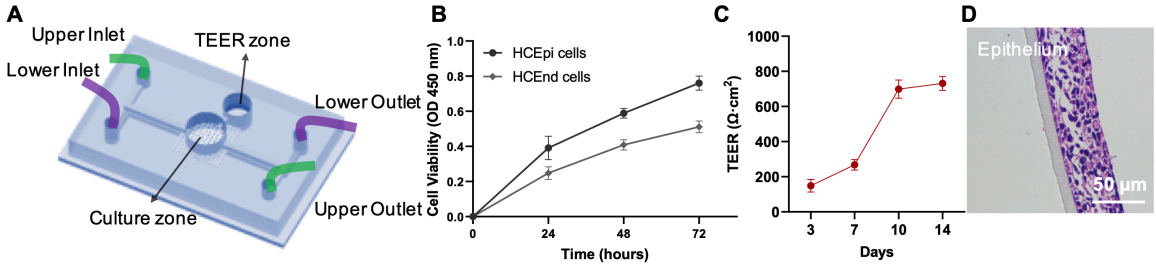

Figure 1. (A) Schematic diagram of the microfluidic-based human cornea-on-a-chip. (B) Cell viability of the HCEpi and HCEnd cells examined by CCK-8 assay. (C) TEER values obtained from the cornea-chip. (D) Representative hematoxylin and eosin images of the stratified corneal epithelium for 14 days. Data are mean ± S.D. from three independent experiments.

Then, a scratch wound is made on the epithelium layer by a pipet tip (Figure 2A). The cultures are photographed at time 0. The wound model is then treated with serum-free cell culture medium as the control group and 1×10^8 MSC-EVs/mL, respectively. After 24 hours, the wounds are again photographed, and the margins are traced. The expression of matrix metallopeptidase 2 (MMP-2) protein was quantitated with a human MMP-2 ELISA kit according to the manufacturer's protocols.

RESULTS AND DISCUSSION

The Cell Counting Kit-8 assay results showed that both HCEpi and HCEnd cells proliferated stably for 72 h, indicating good cell viability (Figure 1B). The TEER values are measured after culturing cells on chip (Figure 1C). The measurement results are presented as $\Omega \bullet cm^2$ by multiplying the resistance with the culture zone's surface area (0.28 cm^2). The chip begins to form a barrier function when the cells are cultured under the ALI condition for 3 days. Besides, the histological section results showed that a stratified epithelium consisting of five to seven stacked cell layers was formed, which recapitulated the corneal epithelium *in vivo* (Figure 1D).

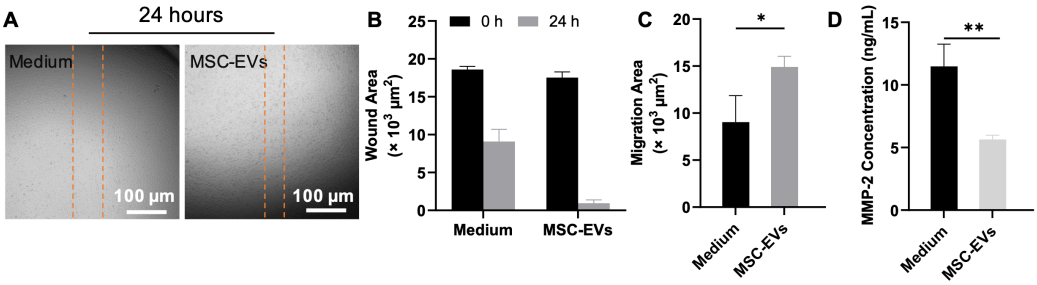

Figure 2. MSCs-EVs stimulate scratch wound healing of human cornea-wound-on-chip. (A) Micrographs showing wound closure at 24 h from different groups. The approximate wound edges are indicated with orange lines. (B) Wound areas at 0 h and 24 h. (C) Migration areas after 24 h. (D) ELISA results of the MMP-2 expression from different treatment groups. All data are mean ± S.D. from three independent experiments. P values by Bonferroni's multiple comparisons test. "" $p < 0.05$, "**" $p < 0.01$.*

At 0 and 24 hours, the approximate wound edges are indicated with yellow lines (Figure 2A). The area between the leading edges of the wound is determined at 24 hours and subtracted from the wound area at time 0 to obtain a value for the wound area (Figure 2B). That of the control group normalizes the results obtained from the experimental groups. MSC-EVs enhance wound healing by approximately 150%, demonstrating that the MSC-EVs are more conducive to cell migration (Figure 2C). The MMP-2 ELISA results showed that the treatment with MSC-EVs was the lowest, indicating that they have the effect of inhibiting corneal inflammation and neovascularization (Figure 2D).

CONCLUSION

We find that MSC-EVs can significantly promote cell migration and reduce the expression of MMP-2 protein, indicating that it could inhibit corneal inflammation and neovascularization, which is beneficial for wound healing. This corneal chip may replace animal models for cornea-related testing, which will provide meaningful, predictive, and complimentary results to *in vivo* and *ex vivo* observations.

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